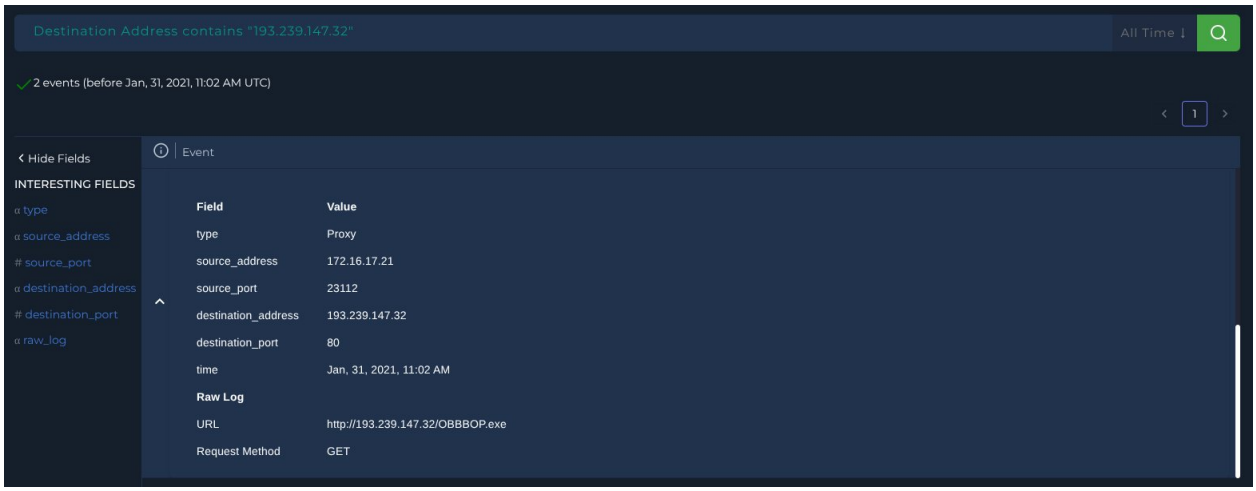
Event Information:

EventID :
43
Event Time :
Jan, 31, 2021, 11:02 AM
Rule :
SOC112 - Traffic to Blacklisted IP
Level :
Security Analyst
Source Address :
172.16.17.21
Source Hostname :
Jack
Destination Address :
193.239.147.32
Username :
jack2021
Request URL :
http://193.239.147.32/OBBBOP.exe
User Agent :
Mozilla/5.0 (Windows NT 10.0; Microsoft Windows 10.0.15063; en-US) PowerShell/6.0.0
Device Action :
Allowed

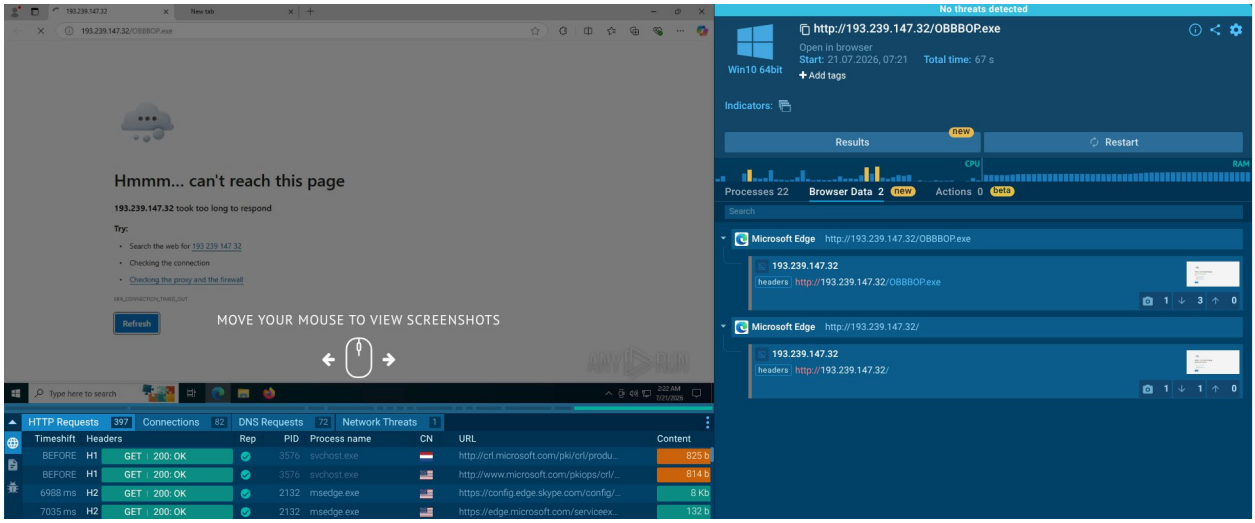
Analysis Steps:

PowerShell being present in the User Agent and the request containing an executable is suspicious.

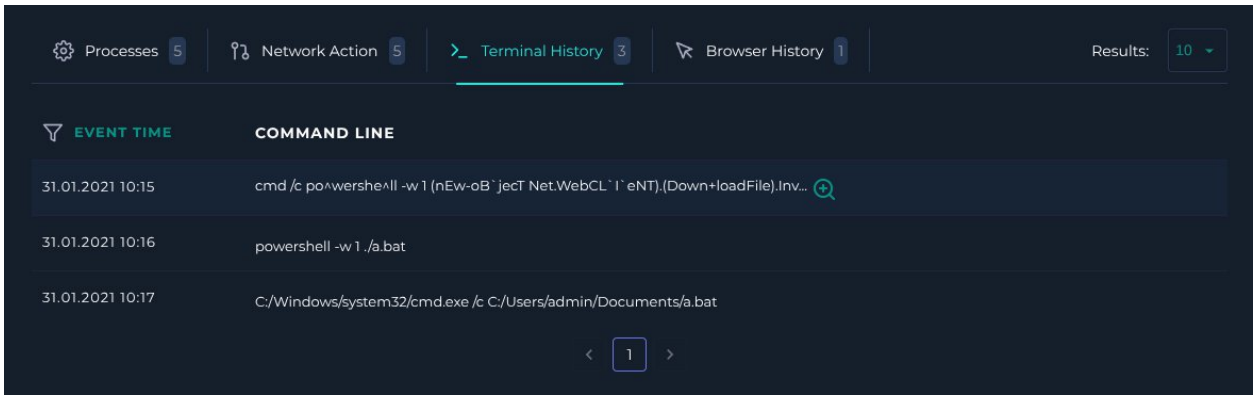
Logs were searched with IP 193[.]239.147.32 as the destination in the relevant time-frame yielding 2 events, we can see the exact event that triggered the alert but not much additional information.



An attempt was made to browse to the same URL in a sandbox but the site seems to be no longer reachable.



The endpoint was analysed to gather additional information as to what might have been going on.



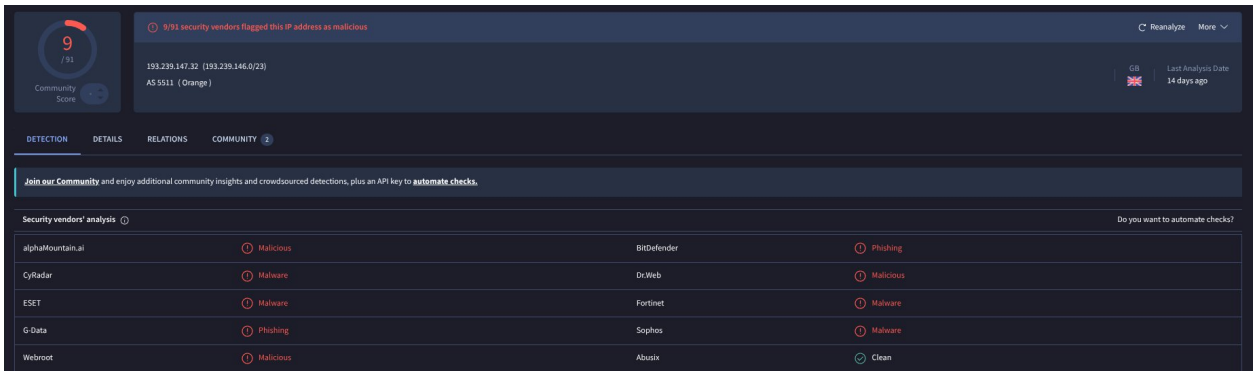
With the available information the steps the attacker likely took are as follows:

Process telemetry indicates that the initial delivery vector likely originated from a macro-enabled Excel document, which subsequently spawned the observed command-line activity. The initial execution at 10:15 utilises the Windows Command Prompt to invoke PowerShell. This command employs caret and backtick characters to obfuscate the execution string, which represents a deliberate evasion technique designed to circumvent static signature-based detection mechanisms. The script instantiates a .NET WebClient object to download a payload from a shortened external URL, designated as hxxp://rebrand[.]ly/WdBPApoMACRO, and saves it locally as a batch file named a.bat. The inclusion of the -w 1 parameter dictates that the process executes within a hidden window, thereby concealing the network request from the end user.

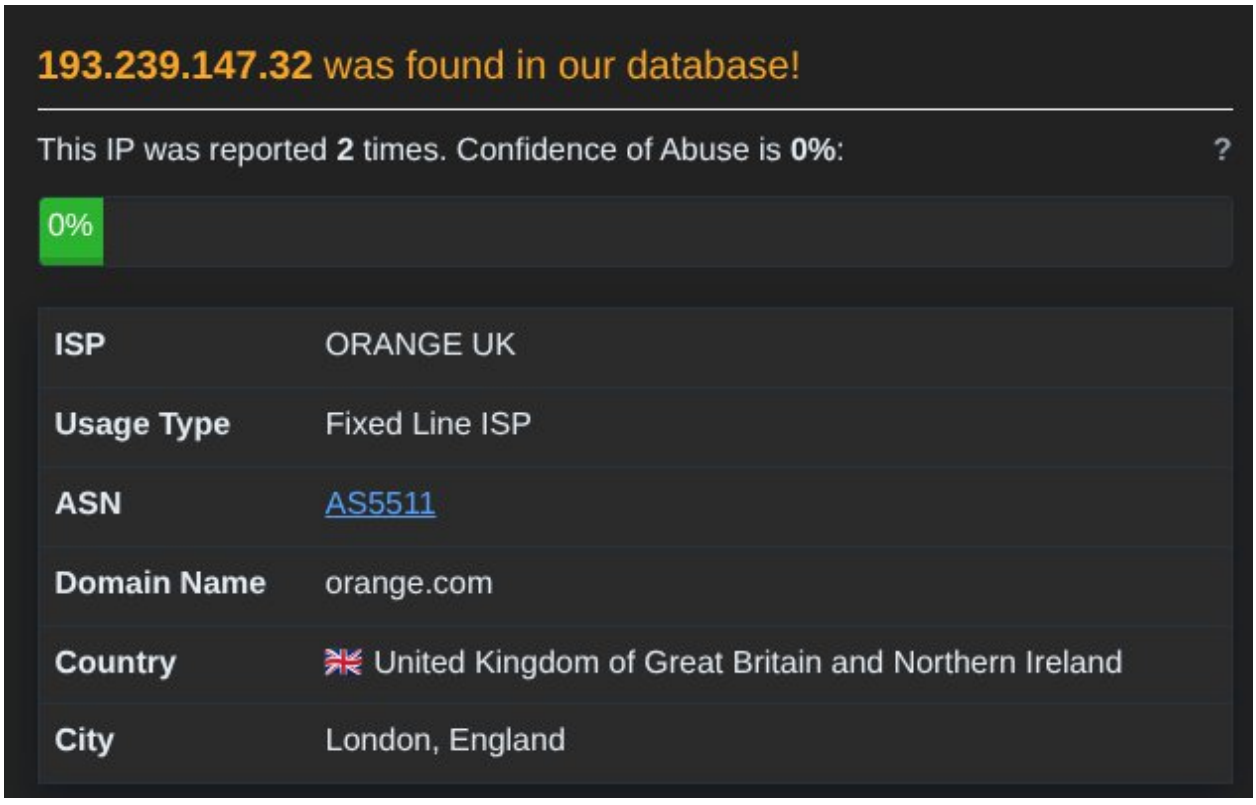
Following this initial download, the telemetry indicates sequential execution of the acquired payload. At 10:16, PowerShell is utilised to execute the batch file directly, again employing the hidden window parameter to maintain operational stealth. Subsequently, at 10:17, the command interpreter is explicitly invoked via its absolute path to execute the identical batch file originating from the user's Documents directory.

This deliberate, repeated execution pattern firmly corroborates a staged deployment strategy occurring directly on the endpoint, ultimately culminating in the network event that triggered the current alert. Specifically, this staged execution facilitated the subsequent outbound connection at 11:02 to the blacklisted infrastructure at 193[.]239.147.32 to retrieve the secondary payload, OBBBOP.exe.

The offending IP is not in our CTI database and is external to the organisation. No emails have been received indicating penetration testing involving the IP for the relevant time period. VirusTotal indicates it is a UK address associated with 193[.]239.147.32 AS 5511 (Orange). 9/91 security vendors flagged this IP address as malicious as of 14 days ago.



On AbuseIPDB this IP was reported 2 times with a 0% confidence of abuse.



Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a True Positive malware infection resulting from staged payload execution. An internal host successfully communicated with a blacklisted external IP address to download a secondary malicious executable. The device action was recorded as allowed, necessitating immediate host containment.

Incident Details

- **Alert Name:** SOC112 - Traffic to Blacklisted IP
- **Target Host:** Jack (172[.]16.17.21)
- **User Account:** jack2021
- **Destination IP:** 193[.]239.147.32
- **Request URL:** hxxp://193[.]239.147.32/OBBBOP.exe
- **Device Action:** Allowed

Analysis and Evidence

- **Initial Delivery:** Process telemetry indicates the initial delivery vector likely originated from a macro-enabled Excel document.
- **Obfuscated Execution:** At 10:15 AM, the Windows Command Prompt invoked PowerShell using caret and backtick characters to obfuscate the execution string. This evasion technique is designed to bypass static signature detection.
- **Payload Acquisition:** The script instantiated a .NET WebClient object to download a payload from hxxp://rebrand[.]ly/WdBPApoMACRO, saving it locally as a.bat. A -w 1 parameter was used to execute this in a hidden window, concealing it from the user.
- **Staged Deployment:** At 10:16 AM, PowerShell executed the batch file directly. Subsequently, at 10:17 AM, the command interpreter (cmd.exe) was explicitly invoked to execute the same batch file from the user's Documents directory.
- **Secondary Payload:** This repeated execution facilitated an outbound connection at 11:02 AM to the blacklisted infrastructure (193[.]239.147.32) to retrieve a secondary payload, OBBBOP.exe.

Threat Intelligence

- **Infrastructure OSINT:** The destination IP 193[.]239.147.32 maps to AS 5511 (Orange) located in the United Kingdom.
- **IP Reputation:** VirusTotal reports that 9/91 security vendors flag this IP address as malicious. AbuseIPDB records 2 historical reports with a 0% confidence of abuse.
- **Internal Correlation:** The offending IP is external to the organisation and is not present in the internal CTI database. No emails have been received indicating authorised penetration testing involving this infrastructure. Sandbox analysis indicates the hosting URL is currently unreachable.

Actions Taken

1. Analysed proxy and terminal history logs to reconstruct the staged execution and payload delivery sequence.
2. Conducted OSINT analysis on the destination IP, confirming its malicious reputation.
3. Attempted dynamic analysis in a sandbox, determining the malicious hosting URL is currently offline.
4. Contained the endpoint from the network.
5. Declared the alert a True Positive and escalated.

Let's Defend Result

